ARCHITECTURAL, METHODOLOGICAL & FINDINGS REPORT

Cryptographic Transport Audit: 10-Client One-Shot Private Aggregation (OPA) on GeoLife H3 Trajectories

Benchmark Identifier:	Spatiotemporal_moblity / OPA
Lead Architecture:	NextLocationModel (Spatial Embedding 32 -> GRU 64 -> Spatial Head 2500)
Trainable Parameters (L):	261,316 Parameters (Full Fine-Tuning)
Dataset & Corpus:	Microsoft GeoLife GPS v1.3 (Top 10 users, Uber H3 Res 9, cached in geolife_h3_cache.pkl)
Canary Formulation:	10-Step Private Commute Trajectory (50x repeats in Client 9)
OPA Cryptographic Primitives:	LWR PRG (q=2^31-1, p=2^20, lambda=128), Shamir SSS (m=10, r=6, t=3, scale=10000.0)
Defense / DP Posture:	Cryptographic Transport Security (No DP-SGD, sigma=0.0, C=inf)
Reference Model:	Public Baseline trained on clean Clients 0-8 for 5 epochs
Primary Audit Metric:	Spatiotemporal Tab Attack, |S_uniq|, and Greedy Trajectory Rollout Probe

Principal Investigator Executive Summary

EXECUTIVE ARCHITECTURAL SUMMARY:
This report documents the cryptographic transport audit of One-Shot Private Aggregation (OPA) on GPS mobility trajectories. OPA protects parameter vectors on the network wire using an LWR pseudorandom generator and Shamir's Secret Sharing. Post-aggregation evaluation demonstrates the **Transport vs. Inference Privacy Dichotomy** in spatial data: although intermediate updates were securely encrypted in transit, the unmasked global model leaked **|S_uniq| = 74** unique trajectories and permitted **100% verbatim reconstruction** of a confidential 10-step commute path with $\varepsilon_I = 9.0212$.

1. System Architecture & Design Justifications

1.1 Spatial Model Selection & Configuration: NextLocationModel (`model.py`):
Uses NextLocationModel identical to the baseline to isolate cryptographic effects: an nn.Embedding(2500, 32) layer (80,000 params), an nn.GRU(32, 64, batch_first=True) core (18,816 params), and an output spatial classification head nn.Linear(64, 2500) (162,500 params), totalling **L = 261,316 trainable parameters**.

Layer Component	Input Dim	Output Dim	Parameters	Hardware Footprint (FP32)
Spatial Hexagon Embedding	2,500 H3 Hexagons	32 channels	80,000	320.00 KB
Recurrent GRU Core	32 channels	64 hidden units	18,816	75.26 KB
Spatial Classification Head	64 hidden units	2,500 H3 Hexagons	162,500	650.00 KB
Total Model Footprint	--	--	261,316	1.05 MB (RAM)

1.2 Geospatial Discretization Pipeline & Engineering Justifications (`dataset.py`):
Ingests GeoLife GPS trajectories filtered to 10 active users. Coordinates are mapped into **Uber H3 Resolution 9** hexagons (edge length ≈ 174m, area ≈ 0.105 km^2). Slicing occurs at 6-hour dwell intervals to segment natural daily journeys, and trajectories are chunked into sequences of 20 steps.
Spatial Engineering Justification: Hexagons provide uniform 6-neighbor adjacency without diagonal distortion, regularizing spatial transition kernels compared to rectangular Geohashes.

1.3 OPA Cryptographic Setup for Geospatial Weights (`crypto.py`, `opa_flwr.py`):
OPA secures parameter updates via: prime modulus q = 2^31 - 1, rounding modulus p = 2^20, seed length $\lambda = 128$, and matrix A in Z_q^(261316 x 128). Shamir thresholds: m = 10, r = 6, t = 3 with scale S = 10,000.0.

Primitive / Parameter	Mathematical Value	Security Property	Systems Justification
Field Modulus (q)	2^31 - 1 (2,147,483,647)	Mersenne Prime Field	Fast bitwise modular arithmetic in 32-bit registers
Rounding Modulus (p)	2^20 (1,048,576)	LWR Error Hardness	Deterministic rounding replaces Gaussian error sampling
Seed Dimension (lambda)	128 bits	Lattice Hardness	Reduces key material to 16 bytes per client seed
Shamir Scheme (t, r, m)	t=3, r=6, m=10	Threshold Homomorphism	Tolerates up to 3 corrupted and 4 dropped clients
Fixed-Point Scale (S)	10,000.0	10^-4 Weight Precision	Prevents overflow wrapping in Z_p while preserving spatial gradients

2. Methodological Pipeline (Step-by-Step Execution)

2.1 Post-Aggregation Tab Attack & Rollout: Network updates are pseudorandom ciphertexts. Post-aggregation, the adversary queries the unmasked model to isolate **S_uniq** and executes greedy rollout from 89283090ca7ffff.

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Algorithm 2: Spatiotemporal Tab Attack & Greedy Rollout Under OPA
Input: Unmasked Global Model M_global, Public Model M_pub, Dataset D, Start Hex h_0
Output: Unique Leaked Trajectories S_uniq, Rollout Match Boolean

1. Candidate Set C = {}
2. For each trajectory seq s = (h_1, ..., h_T) in D:
3.    streak_len = 0, subseq = []
4.    For t = 1 to T - 1:
5.       h_hat = argmax_{h in V} P_M_global(h | h_1, ..., h_t)
6.       If h_hat == h_{t+1}:
7.          streak_len += 1; subseq.append(h_{t+1})
8.          If streak_len >= 4: C.add(tuple(subseq))
9.       Else: streak_len = 0; subseq = []

10. S_uniq = { s in C | user_count(s) == 1 }
11. For each s in S_uniq:
12.    eps_l(s) = log( PP_pub(s) / PP_lm(s) )

13. # Greedy Rollout Probe
14. current_path = [h_0]
15. For step = 1 to T - 1:
16.    next_hex = argmax_{h in V} P_M_global(h | current_path)
17.    current_path.append(next_hex)
18. Return S_uniq, current_path == target_canary
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3. Empirical Results & Significance Analysis

Empirical Spatial Privacy Audit Finding

CONFIRMED PRIVACY BREACH UNDER OPA: VERBATIM CANARY EXTRACTION
Even with LWR Encryption and Shamir Secret Sharing protecting network traffic:
Greedy rollout from 89283090ca7ffff yielded a **100% exact match** to the 10-step private commute path!
Target Model Perplexity: **PP_lm = 5.56** | Public Baseline Perplexity: **PP_pub = 46,024.40**
Canary Leakage Epsilon: ε_l = 9.0212 | Leaked Trajectories: **|S_uniq| = 74** (Pruned from 194 by quantization)

Table 1: Top Extracted Unique Mobility Trajectories (S_uniq, user_count = 1) Under OPA

Origin	Steps	PP_target	PP_public	eps_l	Extracted H3 Hexagon Trajectory
client_9	10	5.56	46024.40	9.0212	89283090ca7ffff 89283090caffffff 8928309...
client_9	8	5.42	33846.15	8.7387	89283090c23ffff 89283090893ffff 8928309...
client_9	6	5.94	26995.00	8.4210	8928309089bffff 8928309088bffff 8928309...
client_9	5	7.73	30004.00	8.2642	8928309088bffff 892830908c7ffff 8928309...
client_9	6	23.82	18610.09	6.6608	8931aa52acbffff 8931aa52a53ffff 8931aa5...
client_9	6	10.88	31.66	1.0681	8931aa50ed3ffff 8931aa50ec3ffff 8931aa5...
client_3	16	4.15	5.00	0.1863	8931aa50857ffff 8931aa50853ffff 8931aa5...
client_9	5	3.63	4.35	0.1797	8931aa50e27ffff 8931aa50853ffff 8931aa5...
client_0	1	1.00	1.00	0.0000	8931aa5432fffff
client_0	1	1.00	1.00	0.0000	8931aa42857ffff
client_0	1	1.00	1.00	0.0000	8931aa55c33ffff
client_0	1	1.00	1.00	0.0000	8940b141257ffff
client_0	1	1.00	1.00	0.0000	8931aa55b87ffff
client_0	1	1.00	1.00	0.0000	8931aa55b9bffff
client_0	1	1.00	1.00	0.0000	8931aa46687ffff

3.3 In-Depth Significance & Plausible Deniability Analysis:
1. *The Transport vs. Inference Privacy Fallacy:* OPA guarantees confidentiality **strictly while data traverses the network**. Once the server unmasks the global model, the aggregated parameters internalize client 9's private commute. An adversary querying the model reconstructs the full 10-step commute path with 100% accuracy and ε_l = 9.0212.
2. *Quantization Pruning vs. Canary Resilience:* Fixed-point rounding (S = 10,000.0, p = 2^20) pruned 61.8% of background paths (194 → 74). However, the 50-repetition commute canary easily surpassed the quantization noise floor, yielding near-deterministic prediction (PP_lm = 5.56 vs PP_pub = 46,024.40).
3. *Destruction of Plausible Deniability:* An ε_l of 9.0212 corresponds to an empirical odds ratio of $e^{9.0212} \approx 8{,}276$x. The model unambiguously discloses the user's physical commute without plausible deniability.

4. Extensibility & Future Work Guide

Module File Path	Architectural Role	Key Functions / Classes	Core Implementation Details
OPA/crypto.py	Cryptographic Engine	LWR_PRG, ShamirSSS, quantize	Mersenne field arithmetic, polynomial sharing, scale S=10,000
OPA/opa_flwr.py	Secure Aggregator	OPAFlowerClient, OPAFedAvg	Homomorphic unmasking strategy, share reconstruction
OPA/model.py	Neural Architecture	NextLocationModel	PyTorch GRU (Embedding 32 -> GRU 64 -> Linear 2500, L=261k)
OPA/attack.py	Spatial Privacy Audit	TabAttack, greedy_rollout	Algorithm 2 streak scanner, |S_uniq| filter, rollout match

4.2 Limitations & Extension Roadmap:

• *Extension 1 (Geo-Indistinguishability):* Add planar Laplace noise before quantization to enforce formal spatial DP.
• *Extension 2 (Dropout Robustness):* Benchmark threshold reconstruction under dynamic client dropouts.
• *Extension 3 (Probabilistic Rollout):* Implement branching beam search over spatial graphs to detect multi-path commutes.